COMPLETE SYSTEM ARCHITECTURE & TECHNOLOGY GUIDE

E-Commerce Platform: Web, Backend & Mobile App

Detailed Step-by-Step Technical Documentation, Module Analysis & Technology Stack

APPLICATION	RUNTIME / NODE	DATABASE	STATUS
Next.js Fullstack + Expo Mobile	Node.js v20+ / React 19 / React Native 0.88	PostgreSQL 16 + Prisma ORM 6.4	Production Ready & Fully Tested

1. System Architecture Overview (High-Level Design)

This project is an enterprise-grade, modern omnichannel e-commerce ecosystem consisting of a **Web Storefront**, an **Advanced Admin Control Panel**, a **Next.js REST API Backend**, and a **Cross-Platform Mobile App (Android/iOS)** built with React Native and Expo.

TIER	CORE TECHNOLOGY	RESPONSIBILITIES & KEY COMPONENTS
Web Frontend	Next.js 16.4 (App Router) React 19.3 Tailwind CSS v4 Lucide React Icons	Customer storefront, catalog browsing, cart, multi-step checkout, user dashboard, and full administrative panel.
Mobile App	React Native 0.88 Expo SDK 58 (Expo Router v4/v5) Zustand v5 AsyncStorage + SecureStore	Native iOS & Android mobile shopping experience, push notifications, offline cart/wishlist sync, animated UI feedback, camera & location integration.
Backend API	Next.js Route Handlers (Edge & Node.js runtime) Jose JWT & Bcryptjs Nodemailer & S3 SDK	RESTful JSON APIs, authentication & RBAC, catalog management, inventory movements, payment gateway processing, push notification dispatch.
Database & Storage	PostgreSQL 16 (Docker) Prisma ORM 6.4 Local JSON Store Fallback AWS S3 / Local Media Assets	Relational data persistence, ACID transactions, relational indexing, media file storage, and automatic schema synchronization.

2. Frontend Technologies (Web Application)

The web client uses the latest Next.js 16 App Router architecture with React 19 Server Components (RSC) and Client Components for maximum performance, SEO, and interactivity.

TECHNOLOGY	VERSION	USAGE IN PROJECT
Next.js App Router	`16.4.2`	Powers file-based routing (`src/app/`), Server-Side Rendering (SSR), Server Actions, dynamic routes (`[slug]` , `[id]`), and layout nesting.
React	`19.3.0`	Core UI component library utilizing modern hooks (`useState` , `useEffect` , `useRef` , `useMemo`), Concurrent rendering, and transitions.
Tailwind CSS	`v4.0.0`	Utility-first CSS engine with Turbopack integration (`@tailwindcss/turbopack`), modern CSS variables, responsive breakpoints, and dark mode support.
Lucide React	`1.61.0`	Modern, tree-shakeable SVG icon collection used across navigation bars, badges, actions, and administrative tables.
Paytm Checksum	`1.5.1`	Cryptographic signature verification and transaction hash generation for Indian payment gateway checkout flows.

3. Backend & API Architecture (Next.js Server)

The backend runs directly inside the Next.js unified framework using Route Handlers located under `src/app/api/`. It connects to PostgreSQL via Prisma ORM with an integrated local database abstraction layer (`DatabaseStore`) for bulletproof failover.

Key Technologies & Libraries

- **Next.js Route Handlers (HTTP Server)**: Supports standard HTTP verbs (`GET`, `POST`, `PATCH`, `PUT`, `DELETE`) with structured streaming and JSON responses.
- **Prisma ORM (v6.4.1)**: Type-safe database client communicating with PostgreSQL on port 5432. Auto-generates TypeScript types matching schema models.
- **Jose (v6.2.13) & Bcryptjs (v3.0.3)**: Secure JWT generation, verification, and salted password hashing without native binary dependencies.
- **Nodemailer (v10.2.1)**: SMTP email delivery engine for account registration, order confirmation receipts, and low-stock administrative alerts.
- **AWS SDK S3 Client (v3.1158.0)**: Cloud object storage integration for product imagery, variant photos, and marketing banner assets.
- **Expo Push API**: HTTP/2 push notification dispatcher delivering real-time push alerts to mobile devices via `https://exp.host/--/api/v2/push/send`.

API Response Convention

All endpoints utilize a strict, consistent JSON envelope format for predictability:

`{ "success": true, "data": [...], "message": "Operation successful." }` (Success)
`{ "success": false, "error": "Descriptive error message.", "status": 400 }` (Error)

4. Mobile App Architecture (React Native & Expo)

Located in the `mobile/` directory, the mobile app provides an ultra-fast, Flipkart/Amazon-style mobile commerce experience for iOS and Android.

TECHNOLOGY	VERSION	PURPOSE & USAGE
React Native	`0.88.1`	Native cross-platform mobile component runtime compiling to real iOS UIKit and Android Views.
Expo SDK	`~58.0.21`	Managed native runtime, dev server (Metro bundler on port 8081), asset management, and native module bridges.
Expo Router	`~58.0.28`	File-based navigation mimicking Next.js conventions: `app/(tabs)/` for bottom navigation and `app/products/[id].tsx` for dynamic routes.

TECHNOLOGY	VERSION	PURPOSE & USAGE
Zustand	`^5.0.16`	Lightweight, hook-based global state management powering: • `cartStore.ts` (Shopping cart state & badges) • `wishlistStore.ts` (Wishlist with self-healing image recovery) • `authStore.ts` (Customer authentication & token persistence) • `toastStore.ts` (Floating animated action notifications) • `notificationStore.ts` (Push & in-app alerts)
AsyncStorage & SecureStore	`v3.1.1 / v58.0.2`	Encrypted storage for JWT session tokens (SecureStore) and persistent offline storage for cart and wishlist items (AsyncStorage).
Expo Notifications	`~58.1.7`	Handles device push token generation, foreground banners, and background notification tap navigation.
Axios Client & Auto-Failover	`^1.21.0`	Custom HTTP networking module with dynamic Wi-Fi LAN IP detection (`10.196.90.130`) and dual-port fallback (3001 & 3000).

5. Step-by-Step Module Breakdown (All 29 Modules)

The codebase is architected into modular domain units located in `src/modules/`. Below is the comprehensive technical breakdown of every module:

MODULE	TECHNOLOGIES	DATABASE MODELS & FILES	FUNCTIONALITY & WORKFLOW
1. Products	Next.js, Prisma, S3 SDK, Lucide	`Product`, `Variant`, `Asset` `src/modules/products/`	SKU/UPC management, multi-warehouse stock tracking, variant matrix (Color, Storage, Size), Google Merchant Center syndication.
2. Categories	React, Tailwind, Prisma	`Category` `src/modules/categories/`	Multi-level category tree (parent-child), slug generation, navigation hierarchy, banner association.
3. Cart	Zustand, React 19, Lucide	`CartItem`, LocalStorage `src/modules/cart/`	Real-time price calculation, quantity adjustments (+/-), stock validation, guest cart to user cart migration upon login.
4. Checkout	Paytm, Razorpay, Tailwind	`Order`, `Address` `src/modules/checkout/`	Shipping address selection, pin code serviceability check, delivery method, payment gateway redirection.
5. Marketing & Push	Expo Push API, Lucide, Tailwind	`Notification`, `PushToken` `src/modules/marketing/`	Broadcast flash deals to mobile devices, master ON/OFF toggle switch, live smartphone mockup preview, deep link routing.
6. Auth & RBAC	Jose JWT, Bcryptjs, Prisma	`User`, `Role`, `Permission` `src/modules/auth/`	Encrypted passwords, role permissions (Admin, Manager, Staff, Customer), JWT verification middleware.
7. Orders	Prisma, PDF generation, Lucide	`Order`, `OrderItem`, `Shipment` `src/modules/orders/`	Order lifecycle state machine (Pending, Paid, Shipped, Delivered, Cancelled), customer order tracking, invoice print.
8. Shipping	Delhivery/BlueDart APIs	`ShippingProvider`, `Zone` `src/modules/shipping/`	Pin-code based zone mapping (Local, Regional, National), COD extra fees, tracking URL generation.
9. Inventory & Warehouses	Prisma, Docker Postgres	`Warehouse`, `InventoryMovement` `src/modules/warehouses/`	Multi-location warehouse stock, low-stock threshold triggers, inter-warehouse stock transfer logging.
10. Reviews & Ratings	React 19, Lucide, Prisma	`Review` `src/modules/reviews/`	1 to 5 star rating system, verified buyer badge, review submission, admin approval toggle.

MODULE	TECHNOLOGIES	DATABASE MODELS & FILES	FUNCTIONALITY & WORKFLOW
11. Customer Q&A	Next.js, Prisma	`Question`, `Answer` `src/modules/qna/`	Customer product inquiry system, community & admin answers, answer moderation.
12. Wishlist	Zustand, AsyncStorage	`WishlistItem` `src/modules/wishlist/`	Heart toggle, self-healing image recovery, move-to-cart automation, mobile-to-web sync.
13. Coupons & Promos	Prisma, Date-fns	`Coupon` `src/modules/coupons/`	Percentage & flat discount codes, minimum spend validation, expiry checks, single-use rules.
14. Sliders & Banners	Image CDN, Touch Swiper	`Slider`, `Banner` `src/modules/sliders/`	Responsive hero banners for desktop & mobile, auto-sliding carousel with pagination dots.
15. Payments	Razorpay, Paytm, COD	`PaymentRecord`, `Refund` `src/modules/payments/`	Online transaction verification, webhook listeners, payment status reconciliation.
16. Digital Assets	AWS S3 SDK, Sharp	`Asset` `src/modules/assets/`	Media library, image uploads, thumbnails generation, primary image assignment.
17. Settings & Mail	Nodemailer, SMTP	`SettingItem`, `MailConfig` `src/modules/settings/`	Store profile settings, currency symbols, SMTP configuration, custom transactional emails.

6. Step-by-Step Data Flow & Execution Workflows

A. Mobile Add to Cart & Checkout Flow

1. User interaction: Customer views a product card or details page (`app/products/[id].tsx`) and taps `Add to Cart` .
2. Store & Animation: `useCartStore.addToCart()` is triggered. Button enters a spinner loading state (`ActivityIndicator`).
3. Toast Dispatch: `useToastStore.showToast()` displays a floating pill: `"Added to Cart! 🛍️"` with a `"View Cart →"` shortcut.
4. API Synchronization: Axios dispatches `POST /api/cart` to Next.js on port 3001. Next.js verifies current stock in PostgreSQL and updates cart.
5. Badge Sync: Root tab bar counter in `app/(tabs)/_layout.tsx` updates immediately without screen re-render.

B. Push Notification Broadcast Flow

1. Mobile Device Registration: When the mobile app starts, `registerForPushNotifications()` calls Expo Push Service to get an `ExponentPushToken` and sends it to `POST /api/notifications/register` .
2. Admin Composition: Admin opens `/admin/marketing/notifications` in web browser, enters title/message, selects emoji and action link, and verifies the live phone preview.
3. Master Switch Validation: System checks `settings.notificationsEnabled` . If ON, proceeds; if OFF, alerts admin that push is paused.
4. Expo Dispatch: Backend compiles chunks of push tokens and posts to Expo push server (`https://exp.host/--/api/v2/push/send`).
5. Native Reception: Customer device receives the notification, plays sound, displays top system alert, and when tapped opens the targeted deep link (e.g. `/(tabs)/deals`).

C. Database Dual-Persistence (Postgres + Local Store)

1. Primary Storage: PostgreSQL running in Docker (container: `ecommerce_postgres` , port: 5432) accessed via Prisma ORM.
2. High Availability Layer: `DatabaseStore` in `src/core/db/store.ts` manages in-memory caching and persistent local JSON synchronization so the entire app operates smoothly even during database migrations or network restarts.

7. Running, Building & Deploying the Project

COMMAND	DIRECTORY	DESCRIPTION
`npm run dev`	Root (`/`)	Starts the Next.js Web Storefront & Backend API on `http://localhost:3001` .
`npx expo start`	`mobile/`	Starts the Expo Metro bundler on port `8081` for Android, iOS, or Expo Go testing.

COMMAND	DIRECTORY	DESCRIPTION
`npm run docker:up`	Root (`/`)	Launches PostgreSQL and database services in background via Docker Compose.
`npm run db:seed`	Root (`/`)	Seeds master product catalog, categories, administrative accounts, and initial warehouses.
`npx tsc --noEmit`	`mobile/` & Root	Validates TypeScript type safety and compiles code with zero warnings or errors.